



CSC8632 Data Science in the Wild (Group Project)

**Executive Report: How to drive Ice Cream Sales in Peak
Season using Data & Insights**

by

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1. Executive Summary

Le Glaçon was tasked with a project to investigate how data can help Velvet Blizzard (VB), a multi-national ice cream brand, sell more ice cream during peak season in France. VB sales have struggled in the French market in recent years, due to a lack of data strategy and awareness. Resolving this was found to be crucial for Logistic Managers and VB executives - our key customers - to make well informed decisions about freezer monitoring, site selection and revenues prediction. Effectively gathering and presenting historical and real-time data streams was deemed essential for informed decision-making. These data streams include tourism [1][2], weather [3], footfall, micro/ macro events, competition and sales trends [4][5].

Our solution is to create a series of data-driven, web-based dashboards integrating IoT data from freezers, revenue figures and external, relevant data points from other impactful data streams such as those mentioned above. Sales forecasting, via time series analysis will be visualized on the dashboards along with a micro-event alerts system powered by generative AI and web scraping. Targeted dashboards will be made for the following user groups: Executives, Regional Managers, Logistic Managers and Shop Owner. Each dashboard's contents will be tailored to the needs of its user.

To produce this solution, data will need to be sourced and collected then stored on the Google Cloud Platform (GCP). Google Collab will be used for data wrangling and sales forecasting model development. We envision that the solution would require a team of data scientists, software developers and UX/UI designers to be created, along with IT specialists for deployment and maintenance.

After employing our dashboards, each VB user group will be able to leverage visualizations, statistics and forecasts to make informed decisions regarding both their short and long-term goals. In addition, VB executives will be able to gain insights into long-term revenue forecasts, allowing them to strategize and allocate resources for impactful future investments.

2. Context

Velvet Blizzard has been underperforming in the French ice cream market, facing multiple challenges such as intensified competition, sluggish sales growth, and inefficient supply chain management. These issues directly impact the company's profitability and weaken its competitive position in the market. Due to a lack of data-driven strategies, the company still relies heavily on experience-based decision-making rather than precise predictive analytics, leading to delayed adjustments that fail to keep pace with market changes. Additionally, inventory management issues have further exacerbated the situation, as inaccurate demand forecasts often result in stockouts or overstocking, increasing logistics costs and supply chain complexity. At the same time, the freezer placement strategy lacks scientific validation, causing some high-demand areas to be underserved, thus limiting market potential. Moreover, the lack of real-time data integration between sales performance, consumer behaviour, and external factors such as weather and seasonal trends has led to missed opportunities in optimizing product distribution and promotional strategies. Consequently, the company urgently needs to

optimize its sales strategies by leveraging data analytics to increase market share, improve supply chain efficiency, and establish a stronger competitive position in France.

These challenges are not unique to Velvet Blizzard, as many fast-moving consumer goods (FMCG) retail companies face similar issues. Across the industry, leading companies have successfully leveraged AI and data-driven technologies to overcome these challenges. In addition, some companies have integrated real-time sales tracking and dynamic pricing models, allowing them to adjust strategies based on consumer behaviour and market fluctuations. These successful cases demonstrate that data-driven optimization strategies are key to maintaining market competitiveness, and Velvet Blizzard must draw insights from these industry leaders to develop a more precise and effective sales strategy.

To achieve this, the company must engage multiple key stakeholders, including internal teams (marketing, sales, supply chain management, and data science) and external partners (consumers, retailers, and data providers). By leveraging data analytics and machine learning, Velvet Blizzard can more accurately forecast market demand, optimize inventory management, and enhance freezer placement efficiency, while also improving promotional efficiency by aligning marketing efforts with real-time consumer trends. Ultimately, these efforts will drive sustainable growth and strengthen the company's position in the French market.

3. Needs

Velvet Blizzard's major challenge is to increase sales during peak season. Therefore, their logistics managers were selected as the customer segment since they are responsible for freezer management, product distribution, and identifying new freezer sites.

Refer to Figure 1: Customer Journey of Logistic Manager, a customer journey map is used to help identify the key pain points across different stages of the process, including production, storage, transportation, distribution, review and analysis, and service. In each stage, we explored the specific jobs, pains, gains, feelings, and efforts to understand the pain points of our customers. Consequently, the customer journey map highlights that both the storage stage and review and analysis stage require the most effort, resulting in the most negative feelings throughout the process. These stages present significant pain points for logistics managers and are crucial areas where improvement can make a substantial difference.

Overseeing the entirety of freezers across France is one of the needs for logistics managers since it allows them to monitor the performance of sales in each area and able to adjust the freezer alignment more easily based on customer demand. Following the distribution process, logistics managers also face difficulties in the review and analysis stage. They need to gather insights from retailers and distributors to identify areas for improvement in both logistics and product offerings. However, the lack of efficient data collection and analysis tools makes it hard to identify critical issues, resulting in lower sales performance. Additionally, the production stage also affects the distribution process. Logistics managers struggle with accurately predicting ice cream demand, which is essential to ensure that production schedules align with actual consumption patterns. Without reliable forecasting tools logistics managers face uncertainty, thus resulting in either overproduction or underproduction.

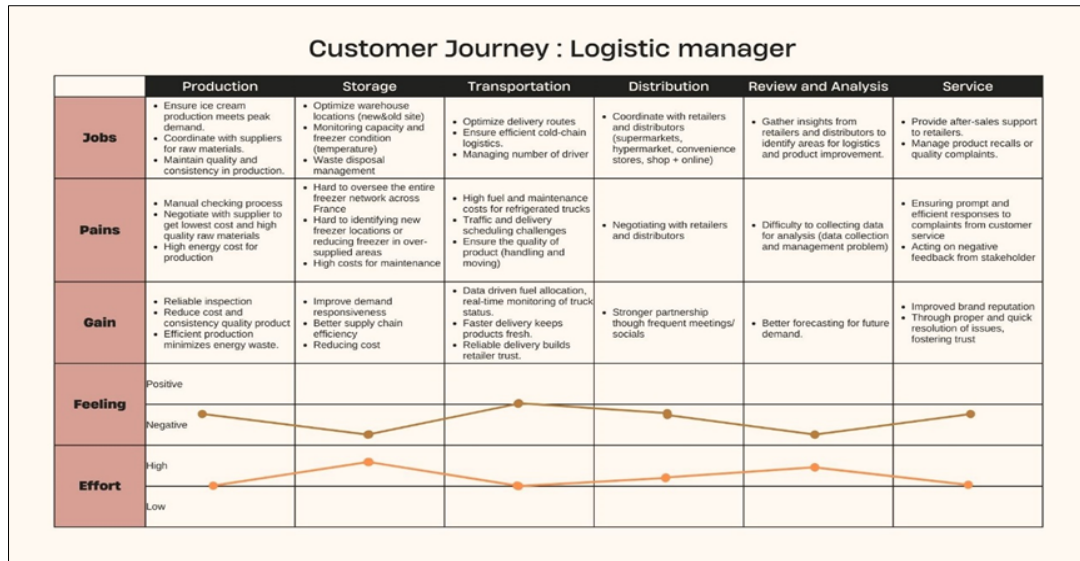


Figure 1: Customer Journey of Logistic Manager

To summarise, there are three major needs for our customer segment including a precision freezer layout system, stock monitoring system, and sales and demand prediction model. However, the customer journey investigation revealed that addressing only one aspect of the problem would not be efficient since the entire customer stages are interconnected. This insight led us to reconsider potential solutions by integrating all needs into one system. Therefore, our team proposes a set of interactive dashboards that include forecasting tools and efficient, explorative data visualization to streamline operations, reduce costs, and improve overall efficiency for logistics managers.

4. Vision

The value proposition canvas in Figure 2: Value Proposition was utilized to analyse the pains and gains of the customers and ensure that our product addresses the pain points of the customer. As a result of the brainstorming session based on the value proposition, pains and tasks were ranked based on the most difficult one to implement. The gains were ranked based on their impact for improving sales (see Figure 3: Ranking of Tasks, Pains and Gains). Therefore, the vision for transforming logistical operations for VB focuses on using an interactive web-based dashboard that combines sales data, IoT data from freezers, and external data sources like weather, travel trends, foot traffic, and local events to address challenges in freezer placement, inventory management, and sales forecasting.

The dashboard tackles key challenges faced by logistics managers, such as preparing for sudden demand, optimizing freezer placement, and analysing sales trends, as highlighted in the Value Proposition canvas (Figure 2: Value Proposition). The solution will help to convert them into gains by anticipating unexpected spikes in demand, predictive modelling will prevent stock shortages and logistical bottlenecks.

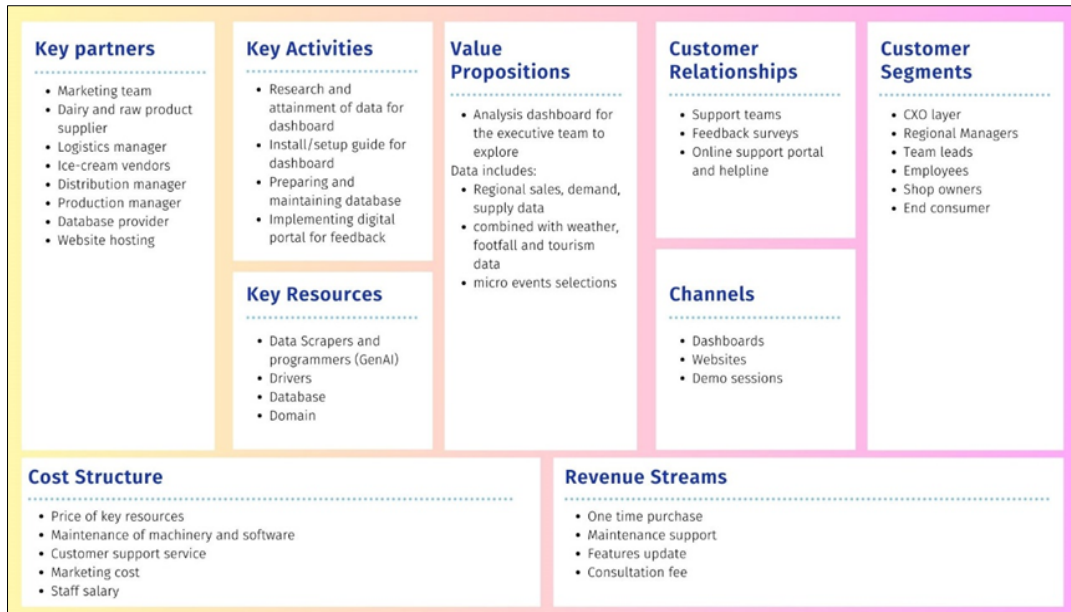


Figure 4: Business Model Canvas

Features

- Freezer Placement Optimization

By utilizing the dashboards, potential freezer locations will be identified using foot traffic, business locations, and historical sales data. Doing so, freezers will be installed at strategic points where there is high demand to maximize sales.

- Sales and Demand Forecasting

Multivariate time series models like SARIMAX and Prophet will predict sales and demand. These models will incorporate key regressors such as weather temperature, weather type (sunny, cloudy, windy), and tourist footfall to improve accuracy. Weekly predictions will align with restocking schedules, and micro-events scraped by Generative AI will be integrated to enhance forecasting precision (see Figure 5: Time-Series Output for Sales Forecast).

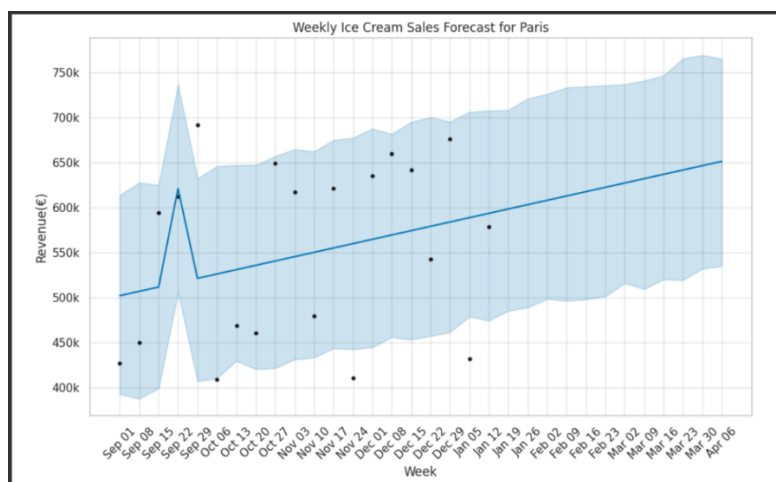


Figure 5: Time-Series Output for Sales Forecast

- Inventory Management

Real-time tracking of stock levels will prevent overstocking, stockouts, and spoilage, ensuring optimal product availability. This will reduce waste and improve operational efficiency, allowing logistics managers to maintain the right balance of inventory across all locations.

- Real-Time Insights

The dashboard will highlight and visualize fluctuations in sales, inventory, and market trends, alerting users when abnormal trends are detected. Role-based access control ensures that executives see national trends, regional managers focus on localized insights, and shop owners receive simplified views of stock levels and upcoming events. Generative AI will scrape micro-events from the internet, feeding them into the dashboard to keep teams informed about local opportunities.

5. Outcome

The data-driven web-based dashboard will be the Minimum Viable Product (MVP). The dashboard will integrate information from the freezer's IoT, revenue and demand forecast and external features like weather, tourist footfall, and upcoming events scraped from the web by Generative AI. This makes the dashboard an efficient tool for the company and shop owners (see Figure 6: Example Dashboard) to monitor performance, estimate the demand for upcoming weeks, and make logistical decisions that will ensure adequate delivery of products.

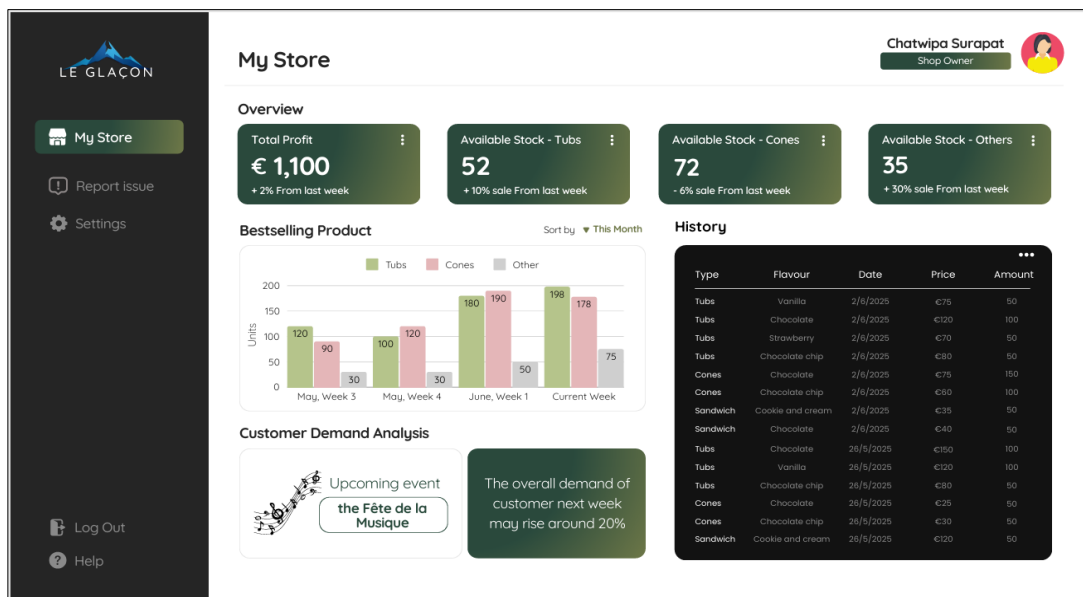


Figure 6: Example Dashboard

The MVP will be developed and deployed by Le Glaçon in coordination with Google's development team. Google's technologies like Google Cloud Platform (GCP) - For data storage, Google Colab - For the time series models and data wrangling, and Gemini (Generative AI) - To scrape information from the web, will be employed to reduce implementation costs. After deployment, ownership will be transferred to Google with occasional product enhancements from Le Glaçon. The success of the MVP will be measured through Key

Performance Indicators such as ice cream sales during peak season and micro-events, the dashboard's ability to predict trends and its impact on the improvement of logistical operations.

Apart from the technical requirements mentioned above, the development of the MVP requires data scientists for the time series models and data analysis, software developers for the dashboard design and for integrating it into the existing system, and UX/UI designers to make the dashboard as intuitive as possible and IT specialists to deploy and maintain the MVP. Along with Google's development team, the MVP requires third-party vendors for additional software tools and services as and when required.

MVP Implementation Plan

The dashboard development roadmap (Figure 7: Dashboard Development Roadmap) for "Le Glaçon" spans six months and is structured into three main phases: **Initial Research**, **Model Development**, and **Product Development**. The process begins with **Business Analysis** in the first month, focusing on identifying business needs and defining functional requirements. In the second month, **Data Wrangling** and **Model Selection & Development** run in parallel, ensuring that data is properly processed and that an appropriate predictive model is chosen. By the third month, efforts shift toward **Model Integration with the Dashboard**, while **Dashboard Design & Development** begins to ensure seamless user interaction.

From the fourth month onward, the focus moves to refining and validating the system. **Internal Testing** is conducted to ensure the dashboard functions as expected, followed by **User Acceptance Testing** in the fifth month to gather feedback from actual users. The **Review & Optimize** phase ensures final refinements and performance improvements before the **Dashboard Deployment** in the sixth month, marking the transition to Google's ownership and long-term maintenance. The roadmap ensures a structured and iterative approach, balancing model development, UI design, and rigorous testing for a smooth deployment.

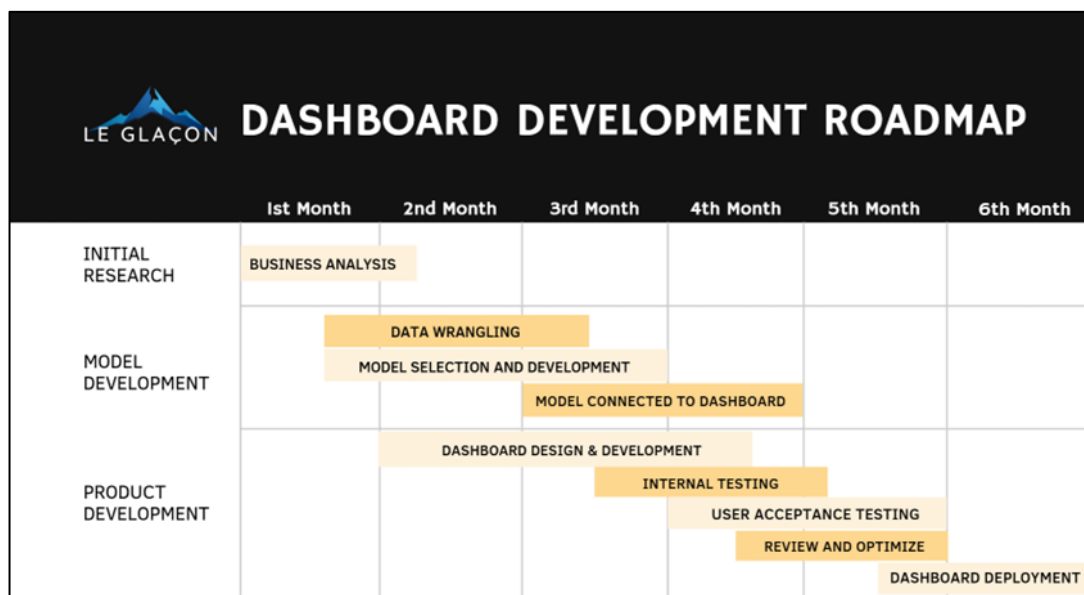


Figure 7: Dashboard Development Roadmap

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